LUDUS LATRUNCULORUM & PETTEIA

Game boards, grammar tables and Greek isopsephy

Summary. Ludus latrunculorum was a Roman two-player board game of capture, believed to be a variant of the Greek games petteia and pessoi. A single remark by Varro compares its grid with the table used to teach noun declensions. The Greek word for a pebble, ψῆφος, gives isopsephy (ἰσοψηφία, "equal pebbles") its name. This note sets out what the sources support, keeps the isopsephy sums apart from the history, and marks every interpretation as such.

I. The name and its ancestry

Latrunculus is a diminutive of *latro*, "mercenary" or "highwayman", so the name means roughly "the game of soldiers" or "of brigands". The game is believed to be a variant of earlier Greek games including *petteia* and *pessoí*. That is a belief, not a proven line of descent. Homer, *Odyssey* 1.107, shows Penelope's suitors playing *pessoí* before the palace doors.

II. Varro and the declension table

Varro, *De Lingua Latina* Book X, is the first Roman mention of latrunculi. He mentions the game in passing, comparing its grid with the table used to set out noun declensions. He does not say the two are identical. Reading the board as a "grammar" is the author's interpretation.

Modern framing (interpretation). In modern linguistics, language runs along a horizontal axis (word order) and a vertical axis (which word is chosen). A board has files and ranks in the same pattern, so the comparison is natural. It is a modern lens and not an ancient claim.

Letter and line. The words γράμμα ("letter") and γραμμή ("line") both come from γράφειν, "to scratch". One is not the root of the other.

III. Classical attestations

- **Ovid, *Ars Amatoria* 3.358:** "unus cum gemino calculus hoste perit", a single counter is lost when caught between two enemies. In *Tristia* 2.477–480 he describes rescuing an isolated counter. The wording there differs ("cum medius gemino calculus hoste perit").
- ***Laus Pisonis*:** a Latin praise-poem of 261 hexameters. The passage on the game is at about lines 190–208, with glass counters (*calculi*, *milites*) mentioned around lines 193 and 196. The date and the identity of the Piso addressed are disputed.
- **Plato, *Republic* 487b:** Socrates compares unskilled debaters to bad *petteia* players, who are finally cornered and unable to move.

IV. What the ground has given

- **Vindolanda (Northumberland):** sixteen latrunculi boards of various sizes have been found at the fort, among them a 7 × 9 board found in 2019.
- **Stanway (Essex):** found in 1996. Kowalski took the board to be 8 × 12 and later allowed 10 × 11. Schädler argues it was neither latrunculi nor 8 × 12.

V. Modern reconstructions of the rules

No ancient rulebook survives, so every set of rules is a modern reconstruction. This table follows the summary in Wikipedia's article "Ludus latrunculorum" and should be checked against the original publications before anyone cites it.

Reconstruction	Board	Pieces
Falkener	12 × 12 (marked uncertain in the source)	five rows per player; exact count not given
Bell (1960)	7 × 8, 8 × 8 and 9 × 10 boards common in Roman Britain	not stated
Bell / Eger-Harsch (1985)	8 × 7	17 per side (16 men plus a Dux)
Kowalski	8 ranks by 12 files; a 10 × 11 variant	12 men plus a Dux each (5 each side of the Dux in the 10 × 11 variant)
Schädler	8 × 8	16 to 24 pieces per player (placed *vagi* and moving *ordinarii*)
Museum Quintana (Künzing)	not given	16 per player; the aim is to capture every opposing piece

VI. The isopsephy

House system: 27 letters (stigma 6, koppa 90, sampi 900), accents ignored, final sigma counted as sigma 200. Every sum below is **computed** and can be checked by adding the letters. Where a sentence reads a meaning into a number, it is marked **model**.

Word	Letters	Sum
Ψηφίο (ψηφίον), little pebble	Ψ 700 + η 8 + φ 500 + ι 10 + ο 70	1288
Ψηφίδιο (ΨΗΦΙΔΙΟ)	Ψ 700 + η 8 + φ 500 + ι 10 + δ 4 + ι 10 + ο 70	1302
Πόλις, city	Π 80 + ο 70 + λ 30 + ι 10 + ς 200	390
Πεττεία, the pebble game	Π 80 + ε 5 + τ 300 + τ 300 + ε 5 + ι 10 + α 1	701
Νόμος, law	N 50 + ο 70 + μ 40 + ο 70 + ς 200	430
Ἀλήθεια, truth	Α 1 + λ 30 + η 8 + θ 9 + ε 5 + ι 10 + α 1	64

Computed: 430 is also the sum of Ἀριθμός (number) and Μαθηματικά (mathematics). An 8 × 8 board has 64 cells, and $64 = 4^3$ = Ἀλήθεια.

Model: in this framework the 64 cells and the 64 of Ἀλήθεια are read together, as a board where a capture follows from position alone. That reading applies only to the 8 × 8 reconstruction (Schädler). Other reconstructions use other boards, so the link is not general.

Not claimed: that any ancient player or author made these links, or that the Greek or Roman games were designed around them.

Sources

- Varro, *De Lingua Latina* X; Ovid, *Ars Amatoria* 3.358 and *Tristia* 2.477–480; *Laus Pisonis*; Plato, *Republic* 487b; Homer, *Odyssey* 1.107.
- Wikipedia, "Ludus latrunculorum" and "Laus Pisonis", used for the archaeological and reconstruction summaries above. Check these against the original publications (Falkener 1892; Bell 1960; Kowalski; Schädler; Richmond 1994 on Laus Pisonis 190–208) before citing.